

I build .NET backend systems where payments, access, and state must survive retries and restarts: explicit state machines, provider re-verification, durable inbox/outbox, idempotency, reconciliation, migrations, and verifiable rollback

Payment correctness

A webhook is not treated as ground truth: provider payment state is re-fetched and checked for id/status/environment/metadata/amount/currency

Recovery ownership

Durable inbox/outbox, idempotency, reconciliation, restart recovery, leases, and manual review for conflicting evidence

Release safety

Versioned releases, coordinated backup, atomic switching, health gates, and rollback to the previous release

EXPERIENCE

- 2026 — present

CompilationLabLMS — payment/backend architecture

 - Made PaymentOrder the owner of allowed state transitions, retry scheduling, and status-check leases; the webhook flow re-fetches payment state from the provider before emitting an internal event
 - Validate payment id/status/environment/metadata/amount/currency and keep malformed or unverifiable provider events separate from confirmed state
- 2026 — present

VpnMediator — state/recovery backend

 - Designed a durable payment inbox/outbox, idempotent reconciliation, and a manual-review path for conflicting provider evidence; restart recovery is covered by dedicated tests
 - Built a versioned release path with coordinated backup, atomic switching, readiness/liveness gates, and restoration of the previous release after failed startup
- 2024 — present

UniversalToolchain — .NET platform architecture

 - Apply the same reliability discipline to platform code through single-owner planning, exact package binding, and regression contracts; the current exact manifest is 1,324/1,324 passing

SKILLS

- NET Backend

C#/.NET, ASP.NET Core, REST/OpenAPI, webhooks, background services, provider integrations
- State & Data

PostgreSQL, SQLite, transactions, explicit state machines, idempotency keys, inbox/outbox, leases
- Reliability

provider re-verification, reconciliation, retry ownership, restart recovery, audit/manual review, backup/restore
- Operations / Platform

Linux, Docker Compose, nginx, systemd, health gates, versioned releases, atomic switching, rollback

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

First-degree diploma and the Grand Prize “Perfection as Hope” for UniversalToolchain

Moscow / remote

SELECTED RESEARCH & ENGINEERING

- CompilationLabLMS

The architecture separates user workflows from financial invariants and supports safe migration and rollback.
- VpnMediator

A production service with controlled state transitions, backup/restore, health gates, and safe rollback; the public case study excludes secrets and user data.
- UniversalToolchain

Current exact test manifest: 1,324/1,324. In a frozen-corpus verifier experiment, B2 detected 32/32 primary and 10/10 challenge operators with 0/100 false positives on valid controls; four post-freeze holdouts were checked separately.